

One quartet, three workflows

Anscombe's data with Knot, R, Python and Typst

1 A reproducible question

Can similar numerical summaries conceal different relationships? Anscombe's quartet gives four small datasets with almost identical means, variances, correlations and least-squares lines, but very different patterns (Anscombe, 1973). This document reproduces that analysis with three workflows.

Variant	Calculation	Drawing
R	All four series in R	ggplot2
Python	All four series in Python	plotnine
Mixed	I–II in R; III–IV in Python	Gribouille

1.1 Data and method

The local file `data/anscombe.csv` contains 44 observations: eleven pairs for each series. It is a long-form transcription of R's `datasets::anscombe`. The values are rounded in the source; “identical” summaries should therefore be understood as approximately equal, not bit-for-bit equal.

For each series we fit $y_i = \alpha + \beta x_i + \varepsilon_i$ by ordinary least squares. We report the sample means, slope, intercept and R^2 . Every figure uses the same axis limits. All regression coefficients and line coordinates are calculated in R or Python before drawing.

1.2 How to read the example

Each variant repeats the question, calculation, numerical summary and graphical interpretation. The `.knot` source contains the executable analysis. Selected loading code is shown here; most computation is hidden for a readable report. The fourth file, `anscombe-errors.knot`, is an opt-in diagnostic companion.

2 R and ggplot2

2.1 Question and data

Do the four series have similar least-squares summaries despite different geometries? Read the same 44 observations from the versioned CSV.

```
1 points <- read.csv("data/anscombe.csv")
```



2.2 Calculation

Estimate a line separately for each series, using sample variances with divisor $n - 1$. Export the unrounded summary for Typst and cross-language checks.

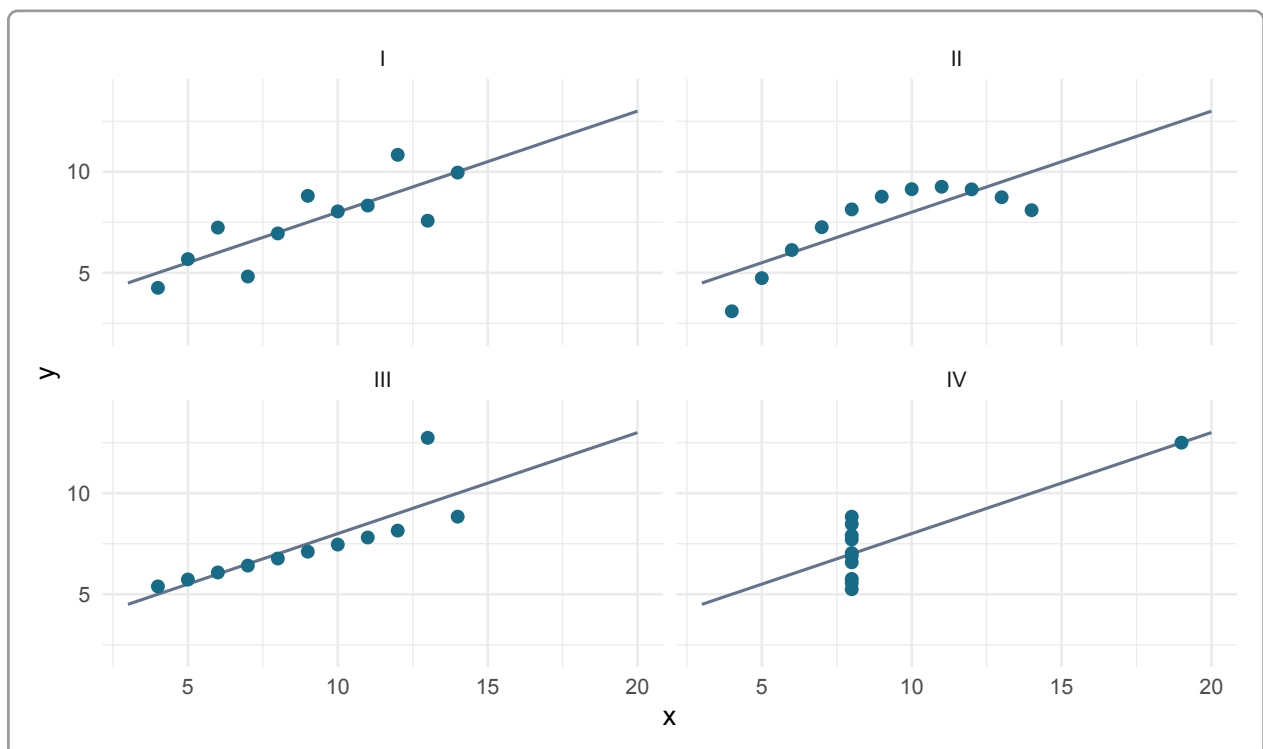
2.3 Numerical summary

Series	Mean x	Mean y	Slope	Intercept	R^2
I	9	7.501	0.5	3	0.667
II	9	7.501	0.5	3.001	0.666
III	9	7.5	0.5	3.002	0.666
IV	9	7.501	0.5	3.002	0.667

There are 44 observations. The means and fitted coefficients are almost the same across the four series. Inspecting only this table would hide the main differences.

2.4 Graphical check

The panels share their scales; the grey lines are the precomputed least-squares fits, not new calculations performed by the graphics package.



What the summaries conceal. Series I has an approximately linear pattern with scatter. Series II is curved. Series III is nearly linear except for an outlying response. In series IV, one observation determines the slope because the other ten share the same x coordinate. Similar fitted lines therefore do not establish that the same model is appropriate for all four datasets.

3 Python and plotnine

3.1 Question and data

Do the four series have similar least-squares summaries despite different geometries? Read the same 44 observations from the versioned CSV.

```
1 import pandas as pd
2 points = pd.read_csv("data/anscombe.csv")
```

Python

3.2 Calculation

Estimate a line separately for each series, using sample variances with divisor $n - 1$. Export the unrounded summary for Typst and cross-language checks.

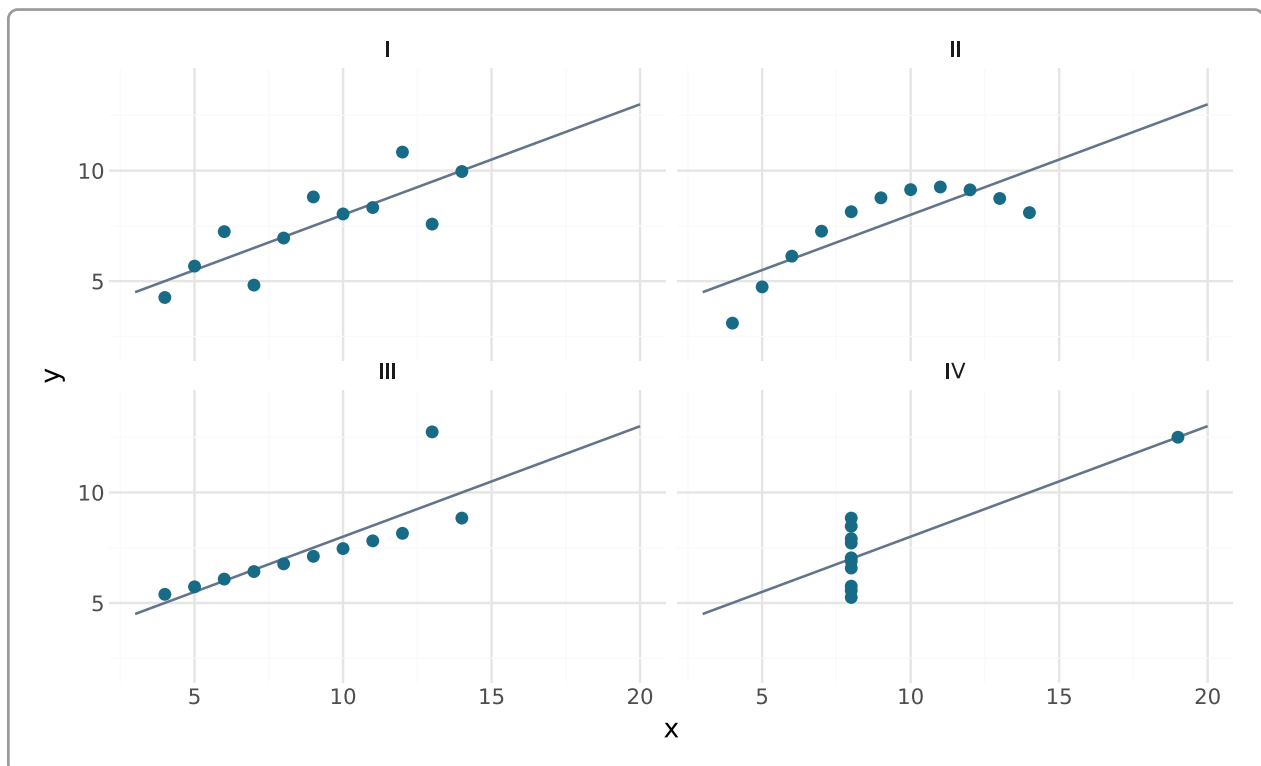
3.3 Numerical summary

Series	Mean x	Mean y	Slope	Intercept	R^2
I	9	7.501	0.5	3	0.667
II	9	7.501	0.5	3.001	0.666
III	9	7.5	0.5	3.002	0.666
IV	9	7.501	0.5	3.002	0.667

There are 44 observations. The means and fitted coefficients are almost the same across the four series. Inspecting only this table would hide the main differences.

3.4 Graphical check

The panels share their scales; the grey lines are the precomputed least-squares fits, not new calculations performed by the graphics package.



What the summaries conceal. Series I has an approximately linear pattern with scatter. Series II is curved. Series III is nearly linear except for an outlying response. In series IV, one observation determines the slope because the other ten share the same x coordinate. Similar fitted lines therefore do not establish that the same model is appropriate for all four datasets.

4 R and Python, drawn by Gribouille

4.1 Question and data

Repeat the same analysis, assigning series I–II to R and III–IV to Python. Both chains read the input independently; the document order does not impose an execution order between languages.

```
1 points <- read.csv("data/anscombe.csv")
2 points <- subset(points, series %in% c("I", "II"))
```

R

```
1 import pandas as pd
2 points = pd.read_csv("data/anscombe.csv")
3 points = points[points["series"].isin(["III", "IV"])]
```

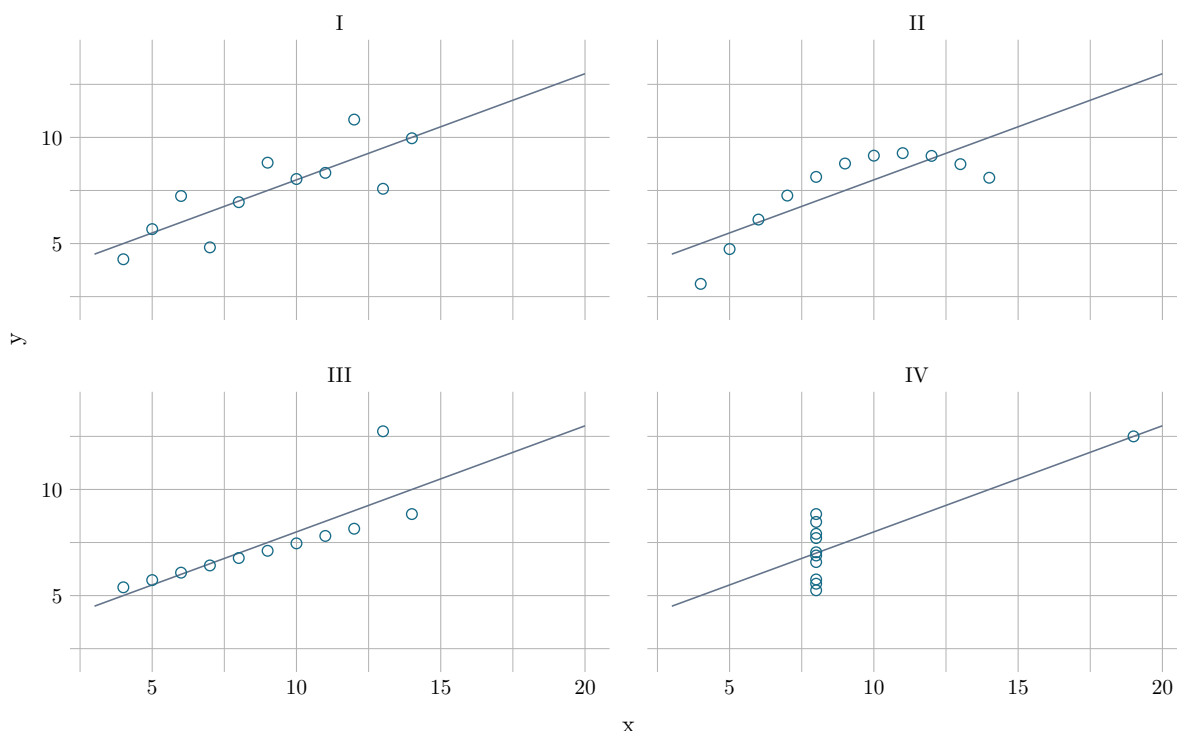
Python

4.2 Numerical summary

Series	Mean x	Mean y	Slope	Intercept	R^2
I	9	7.501	0.5	3	0.667
II	9	7.501	0.5	3.001	0.666
III	9	7.5	0.5	3.002	0.666
IV	9	7.501	0.5	3.002	0.667

R handles 22 observations and Python handles 22. The means, variances and regression coefficients use the same definitions as in the previous variants.

4.3 Graphical check



What the summaries conceal. Series I has an approximately linear pattern with scatter. Series II is curved. Series III is nearly linear except for an outlying response. In series IV, one observation determines the slope because the other ten share the same x coordinate. Similar fitted lines therefore do not establish that the same model is appropriate for all four datasets.

One drawing layer. Gribouille receives the same row structure from both languages. Its role here is graphical composition; it does not estimate a model.

5 Cross-language validation

Verified: R, Python and the mixed workflow agree within an absolute tolerance of 10^{-8} for every reported statistic, including the sample variances. This check executes during PDF compilation.

5.1 Re-execution and scope

Use `knot build` while editing, then `knot build --strict --no-snapshots` for the final validation: a full re-execution that fails on any error. The input CSV is declared as a dependency in every reading chunk. JSON exports and figures are managed by Knot’s cache. Changing prose should reuse calculations; changing the CSV should invalidate them.

The two languages in the mixed file have independent work and may run in parallel. They never read files produced by the other language. Each included `.knot` file has its own interpreter state.

Bibliography

Anscombe, F. J. (1973). Graphs in Statistical Analysis. *The American Statistician*, 27(1), 17–21. <https://doi.org/10.1080/00031305.1973.10478966>